

Relational Interpreters, Program Synthesis, and Barliman: Strange, Beautiful, and Possibly Useful

Will Byrd

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**MINIKANREN IS THE WORST LOGIC PROGRAMMING
LANGUAGE I'VE EVER HEARD OF**



BUT YOU HAVE HEARD OF IT



Credit: Wikipedia

93. When someone says “I want a programming language in which I need only say what I wish done,” give him a lollipop.

–Alan J. Perlis, Epigrams in Programming



Credit: Wikipedia

**Relational Interpreters, Program Synthesis, and
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- or -

Lollipop-Driven Development

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$$\begin{array}{c}
\frac{\text{quote} \notin \text{dom}(\rho)}{\rho \vdash (\text{quote } d) \Downarrow d} \quad (\text{QUOTE}) \\
\\
\frac{\rho \vdash e^* \Downarrow v^* \quad \dots \text{list} \notin \text{dom}(\rho)}{\rho \vdash (\text{list } e^* \dots) \Downarrow v^* \dots} \quad (\text{LIST}) \\
\\
\frac{(x : v) \in \rho}{\rho \vdash x \Downarrow v} \quad (\text{VAR}) \\
\\
\frac{\lambda \notin \text{dom}(\rho)}{\rho \vdash \lambda x.e \Downarrow \langle \lambda x.e \textbf{in } \rho \rangle} \quad (\text{ABS}) \\
\\
\frac{\rho \vdash e_1 \Downarrow \langle \lambda x.e \textbf{in } \rho_1 \rangle \quad \rho \vdash e_2 \Downarrow v_2 \quad \rho_1, (x : v_2) \vdash e \Downarrow v}{\rho \vdash (e_1 e_2) \Downarrow v} \quad (\text{APP})
\end{array}$$

Figure 2: Environment-passing Interpreter
(shadowing allowed)

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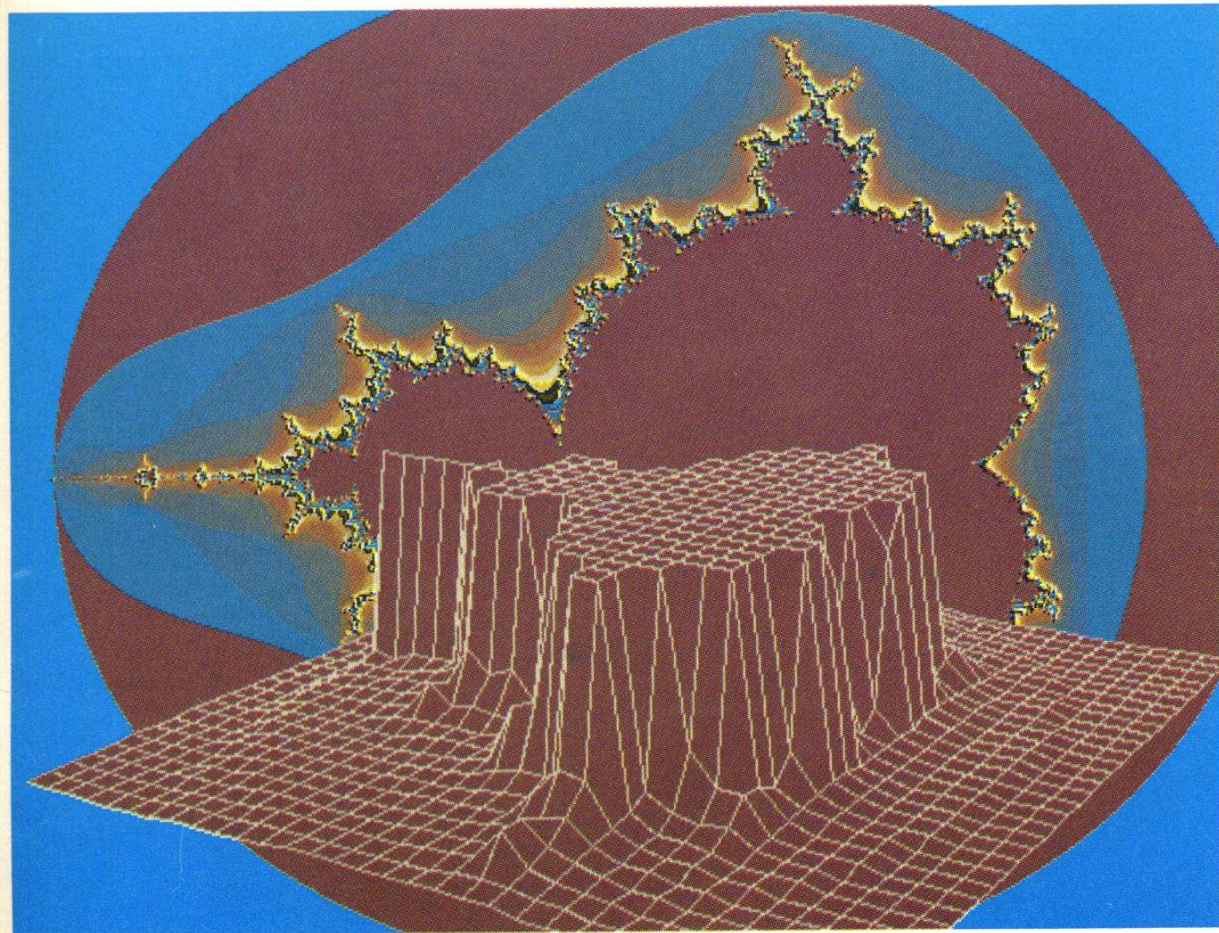
- or -

**miniKanren: An Executable Relational Computer
Science Metanotation**

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Doug's
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AQUARIUM**



Doug Houck
Seven Seas Software
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Credit: Learning Games Initiative

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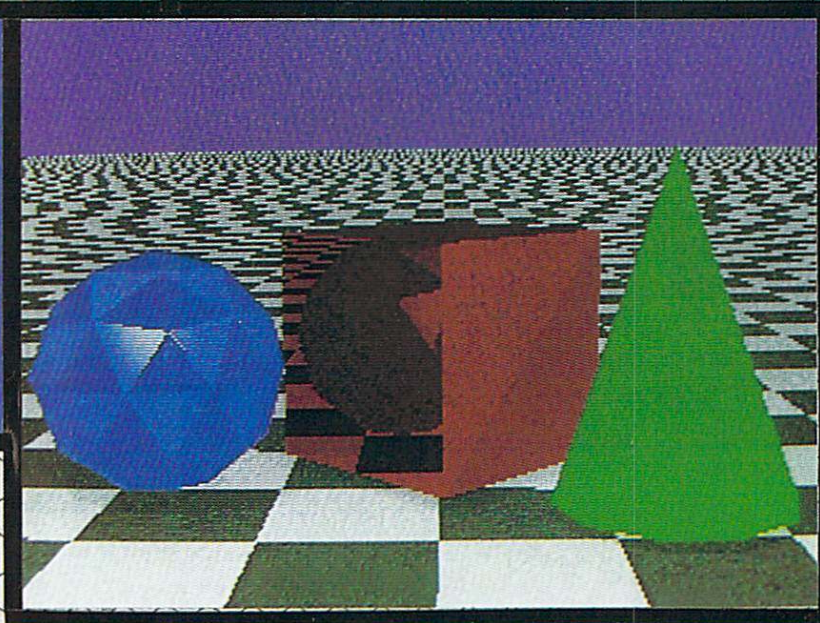
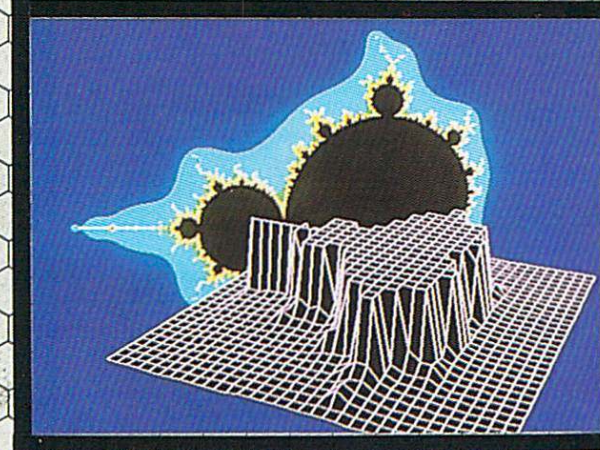
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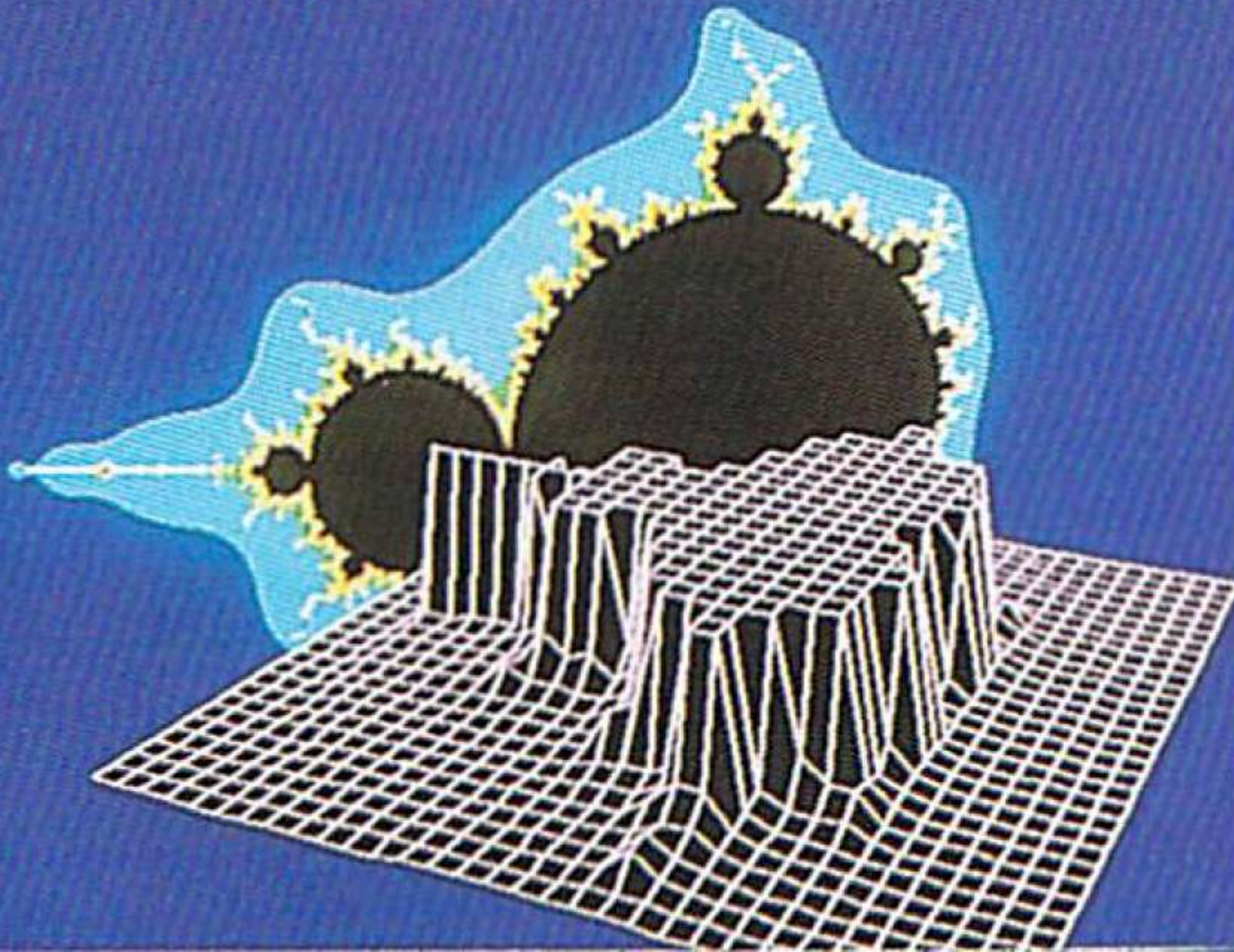
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- New Tools,
- New Toys!





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miniKanren: The Penultimate CSM Aquarium

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